

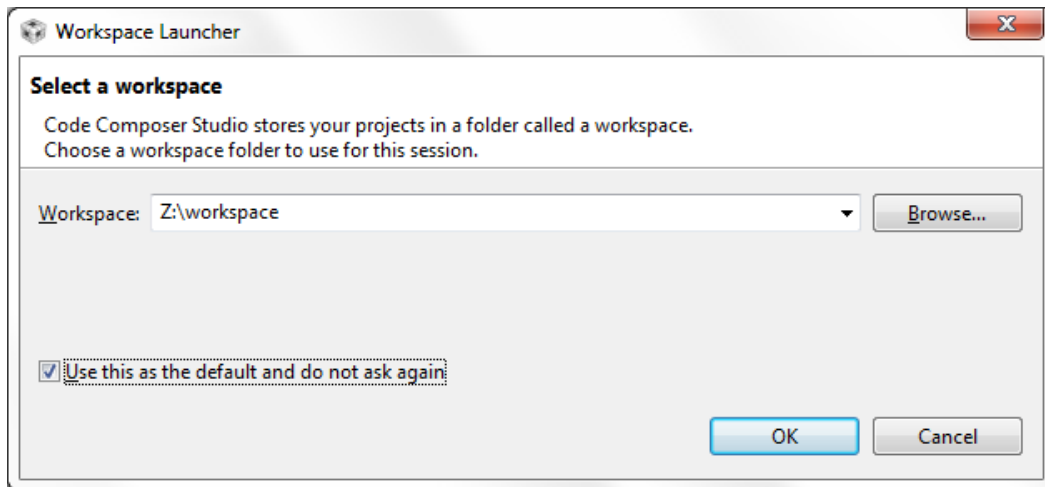
Configuring Code Composer Studio 5 for use with the TMS320C6713 DSK

Tyler Ackland

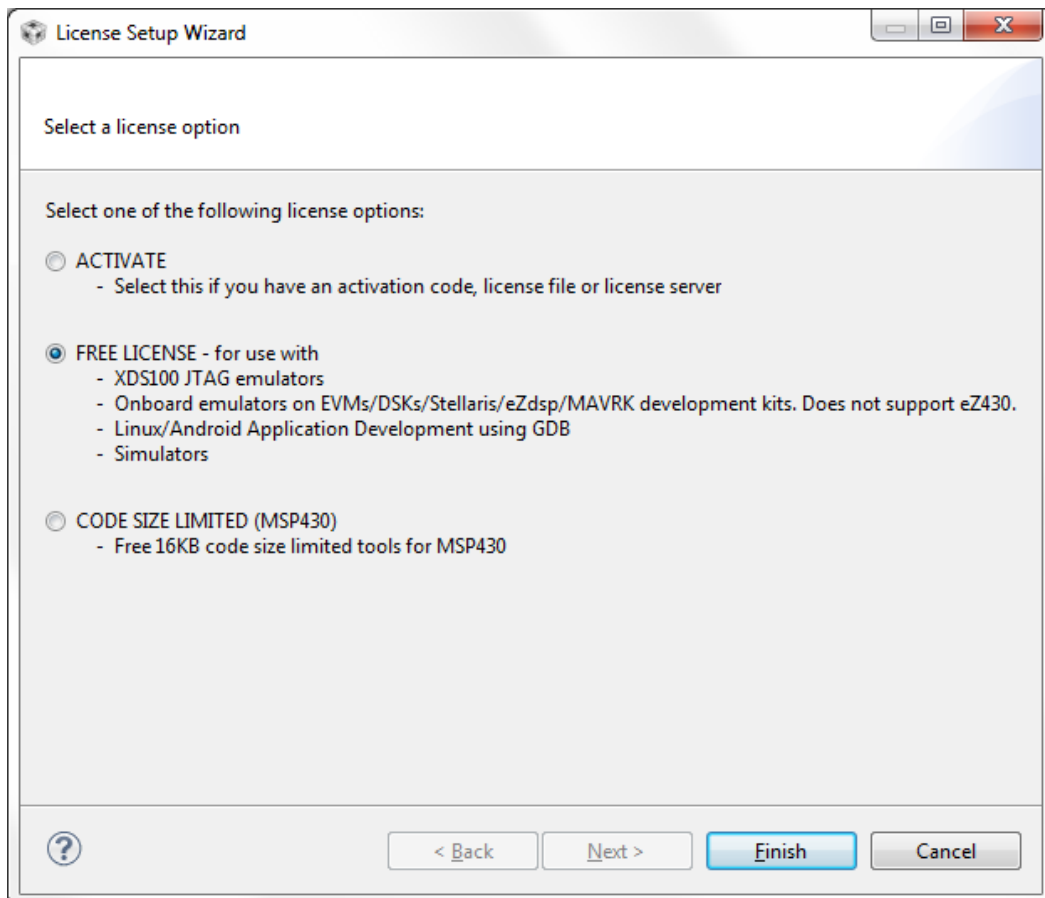
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Part 1: Start Code Composer Studio

- Launch CCS using the desktop or start menu shortcut
- Enter the default workspace as Z:\WORKSPACE



- Select the FREE LICENSE option. The development board activates the software



Part 2: Create the CCS project container

- Pull down the File menu and select "New → CCS Project"
- Type in the Project name and workspace location.
- Select Device Family "C6000" and Variant "TMS320C67XX"
- Set the Connection as "Spectrum Digital DSK-EVM-eZdsp onboard USB Emulator"
- Expand "Advanced settings" and select Linker command file "C6713.cmd"

New CCS Project

Create a new CCS Project.

Project name: xxxxxxxxxxxx

Output type: Executable

☐ Use default location

Location: Z:\workspace Browse...

Device

Family: C6000

Variant: TMS320C67XX DSK6713

Connection: Spectrum Digital DSK-EVM-eZdsp onboard USB Emulator

▼ Advanced settings

Device endianness: little

Compiler version: TI v7.4.1 More...

Output format: legacy COFF

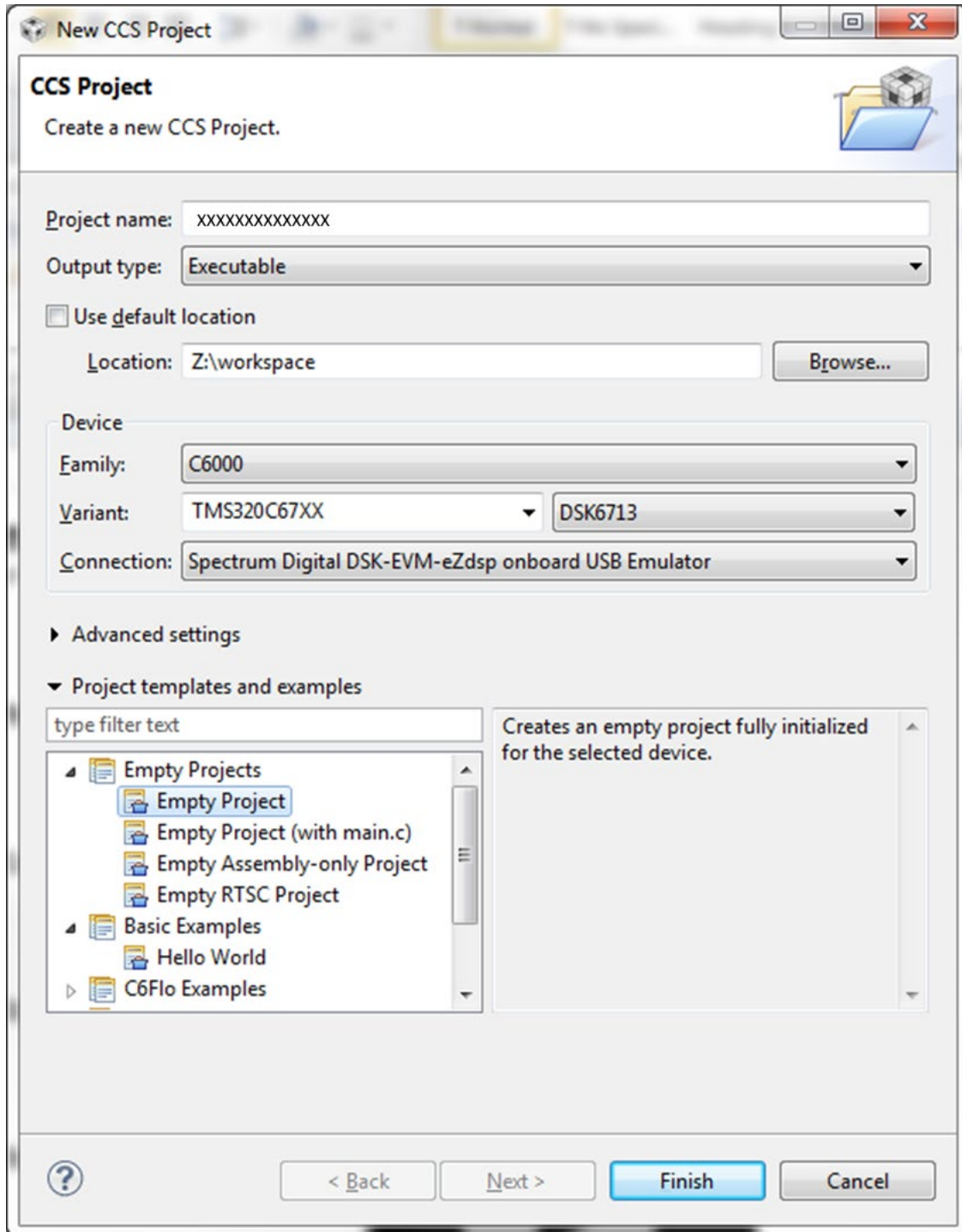
Linker command file: C6713.cmd Browse...

Runtime support library: <automatic> Browse...

► Project templates and examples

? < Back Next > Finish Cancel

- Expand “Project templates and examples” and select “Empty Project”



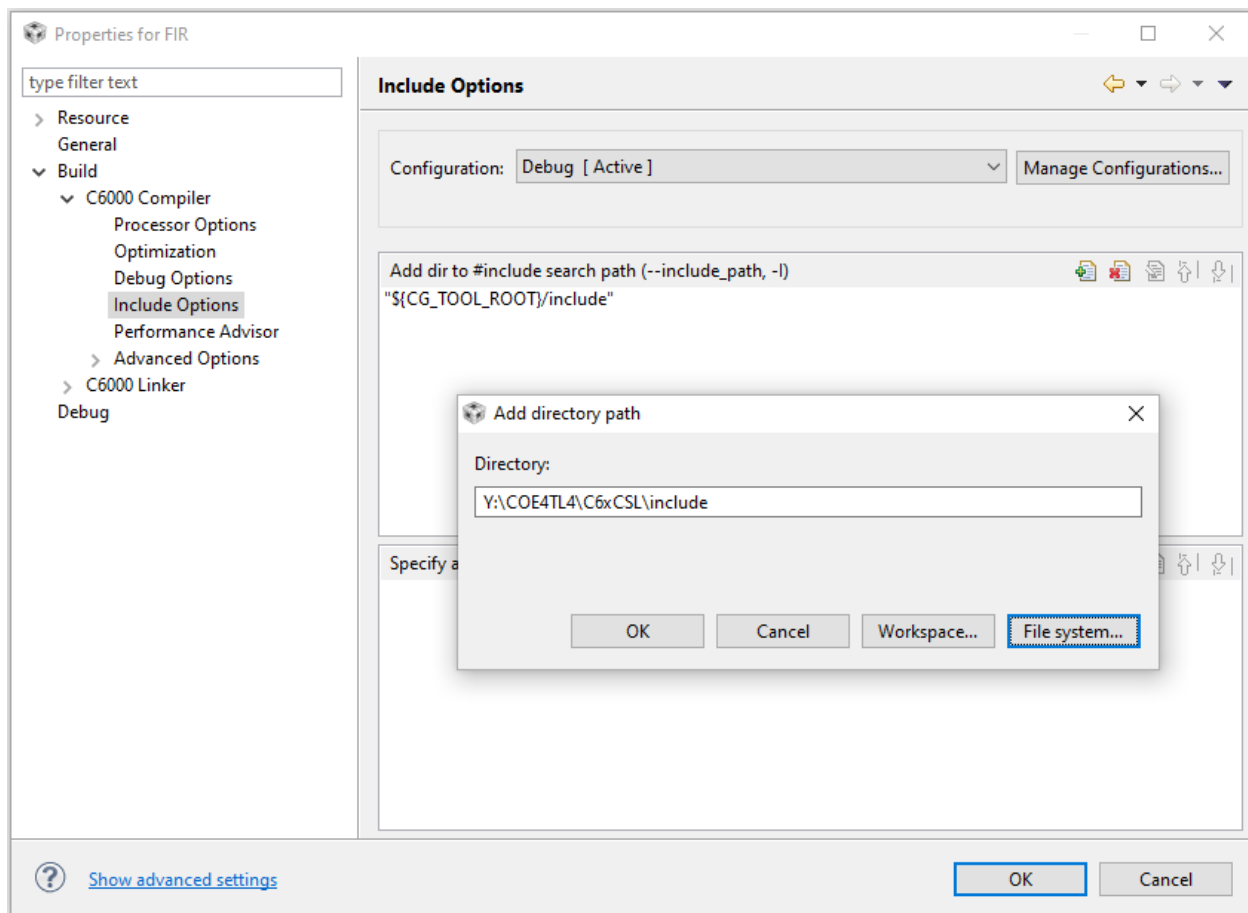
- Click “Finish” to create the project

Part 3: Add source code to the project

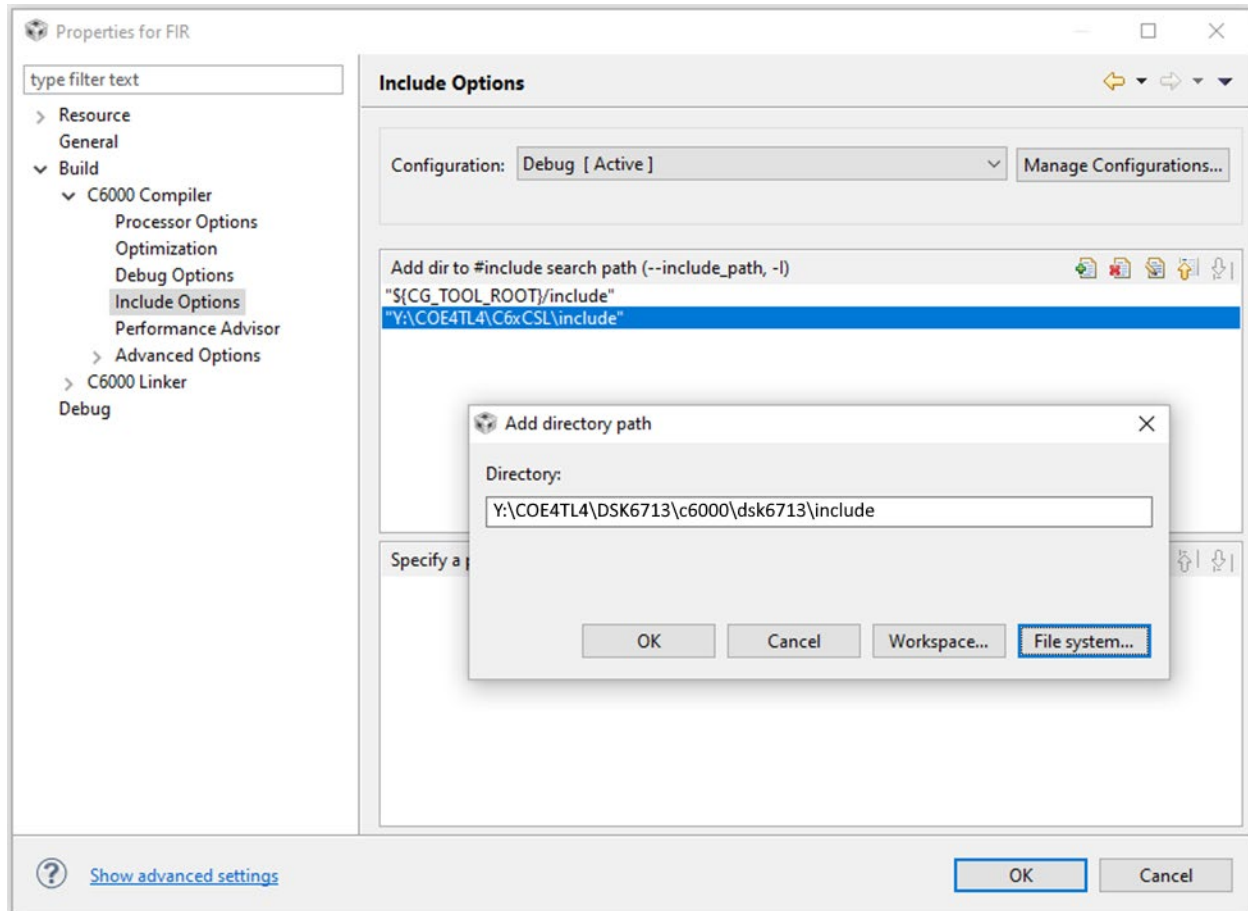
- Pull down the Project menu and select “Add Files...”
- Browse to the directory containing the source code files, which is mentioned in the lab manual.
- Select all .c, .h and .asm files and click open
- Select “Copy files” to copy the files to your project directory

Part 4: Specify compiler include locations

- Pull down the Project menu and select “Properties”
- Expand the Build and C6000 Compiler menus. Select “Include Options”
- Click the icon with the green plus sign to add include locations
- Include the Chip Support Library path supplied by Texas Instruments. It is located at Y:\COE4TL4\C6xCSL\include

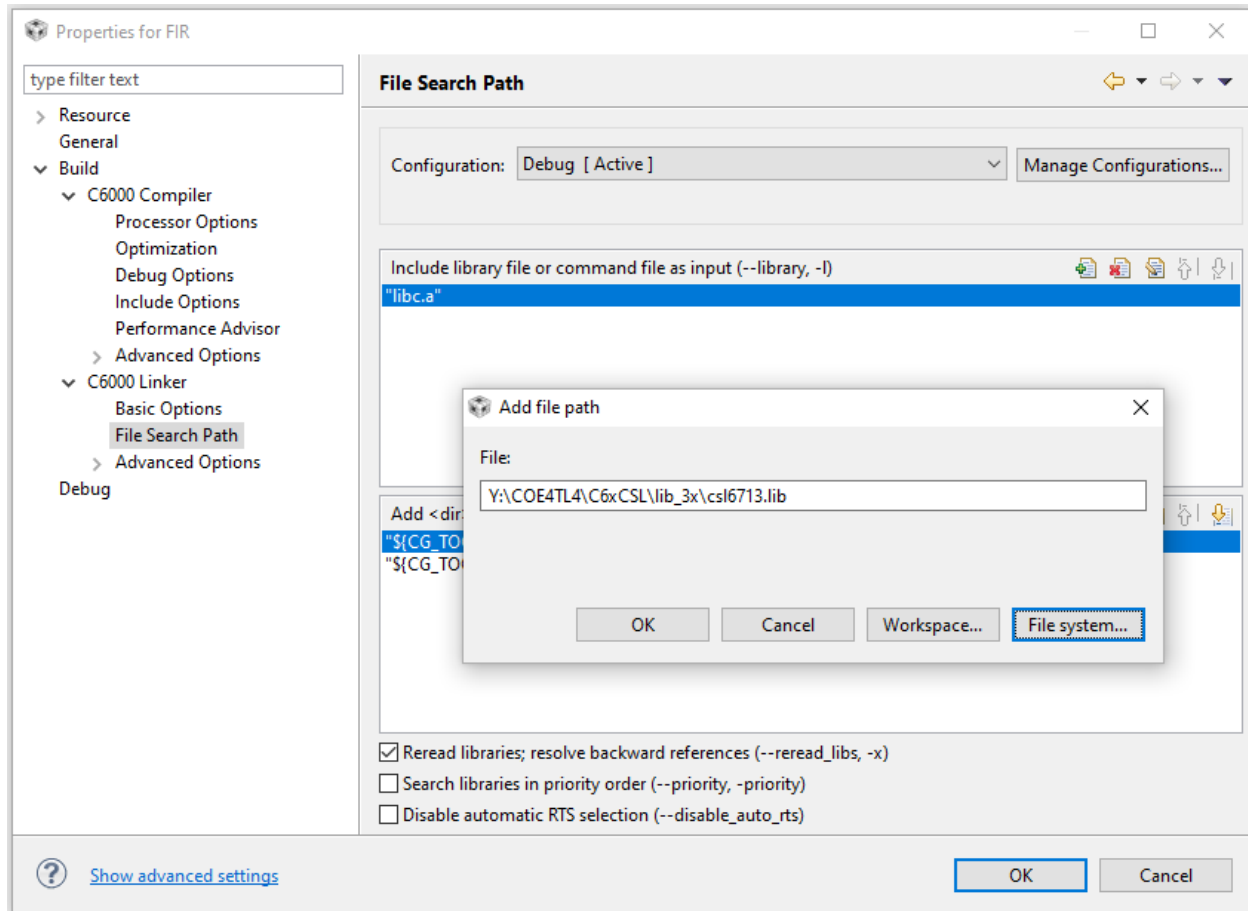


- Include the Board Support Library path supplied by Spectrum Digital. It is located at Y:\COE4TL4\DSK6713\c6000\disk6713\include

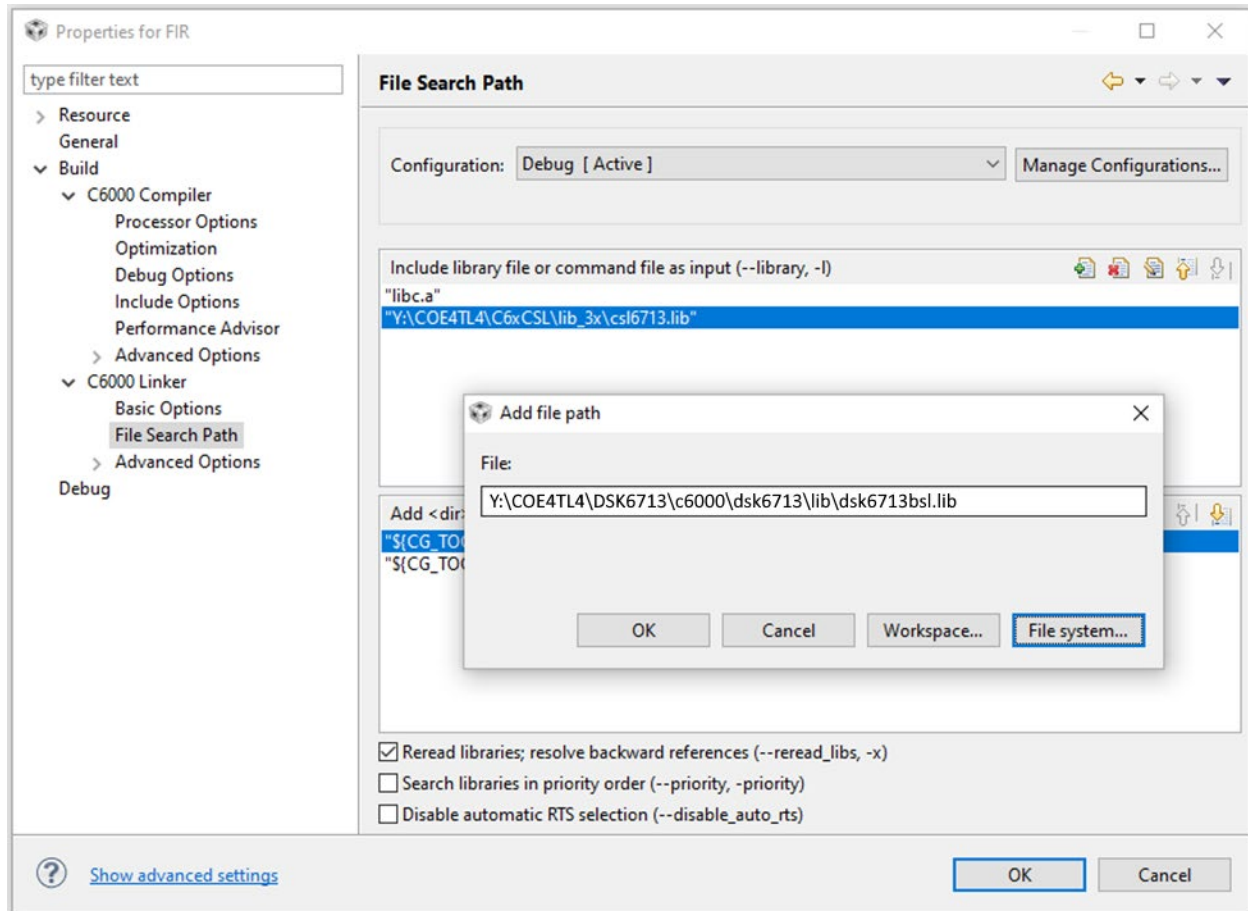


Part 5: Specify linker library locations

- Expand the C6000 Linker menu and select “File Search Path”
- Click the icon with the green plus sign to add library files
- Add the chip support library (CSL) located at Y:\COE4TL4\C6xCSL\lib_3x\cs16713.lib

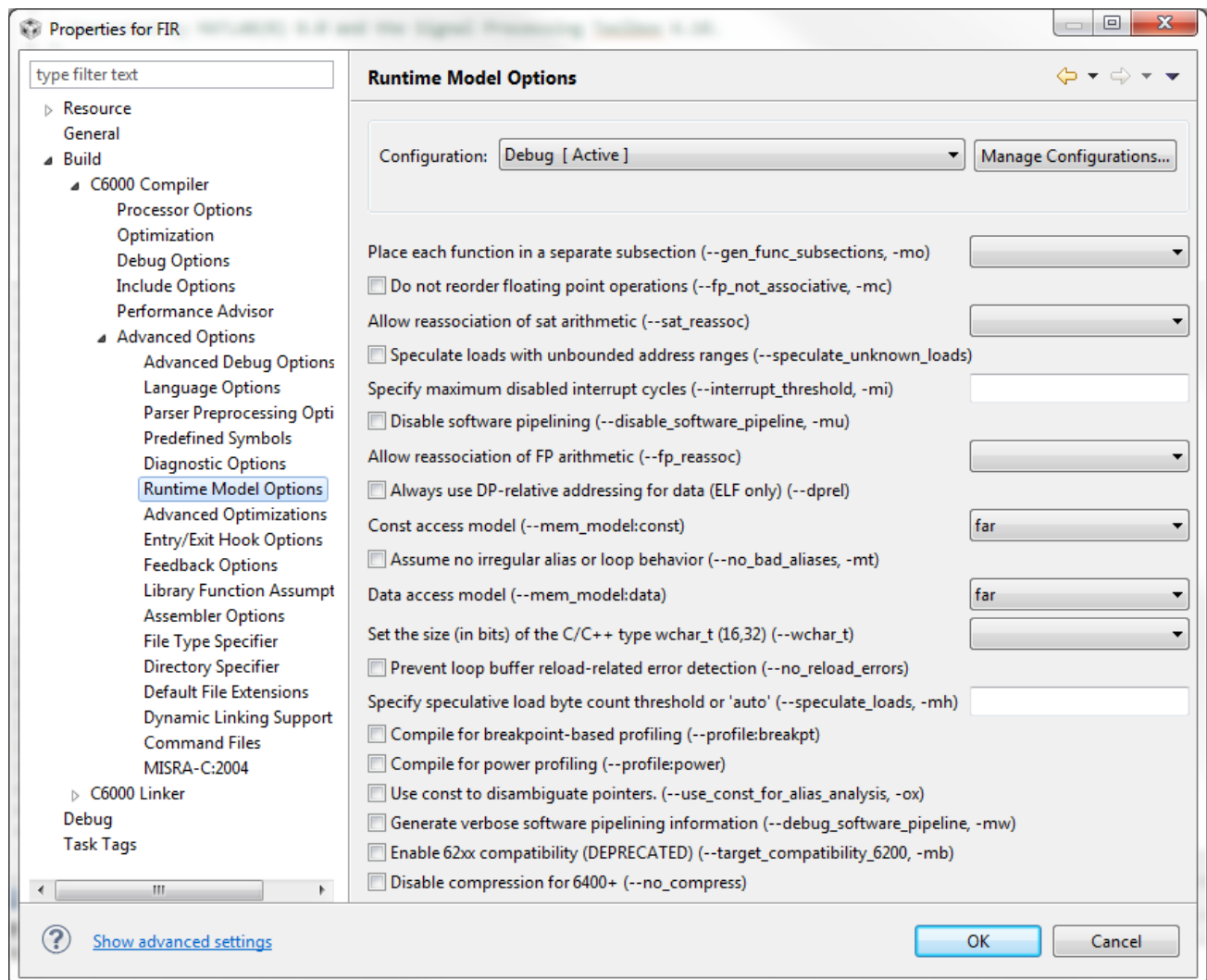


- Add the board support library (BSL) located at
Y:\COE4TL4\DSK6713\c6000\dsk6713\lib\dsk6713bsl.lib



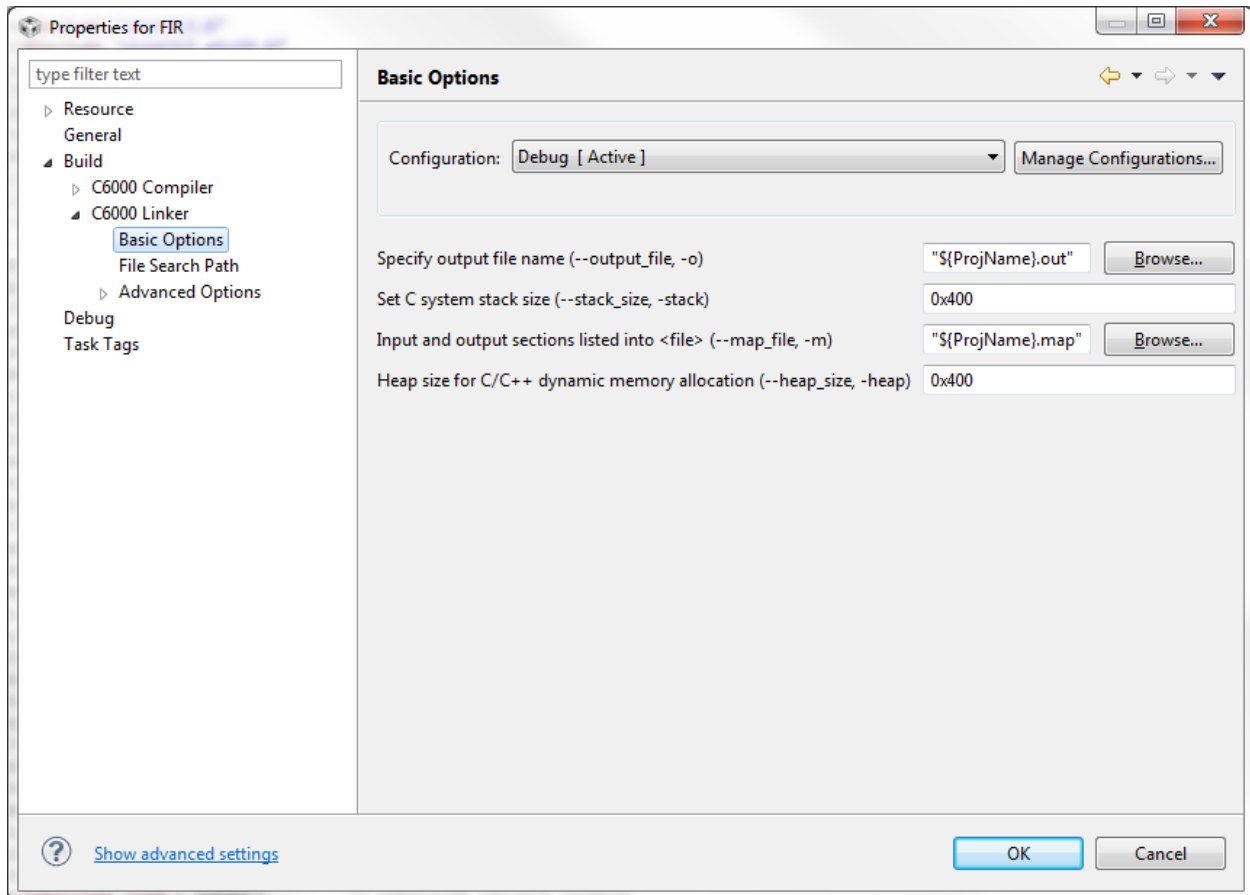
Part 6: Specify "far" memory model

- Expand C6000 Compiler → Advanced Options and select "Runtime Model Option"
- Set the "Const access model" and "Data access model" both to "far"



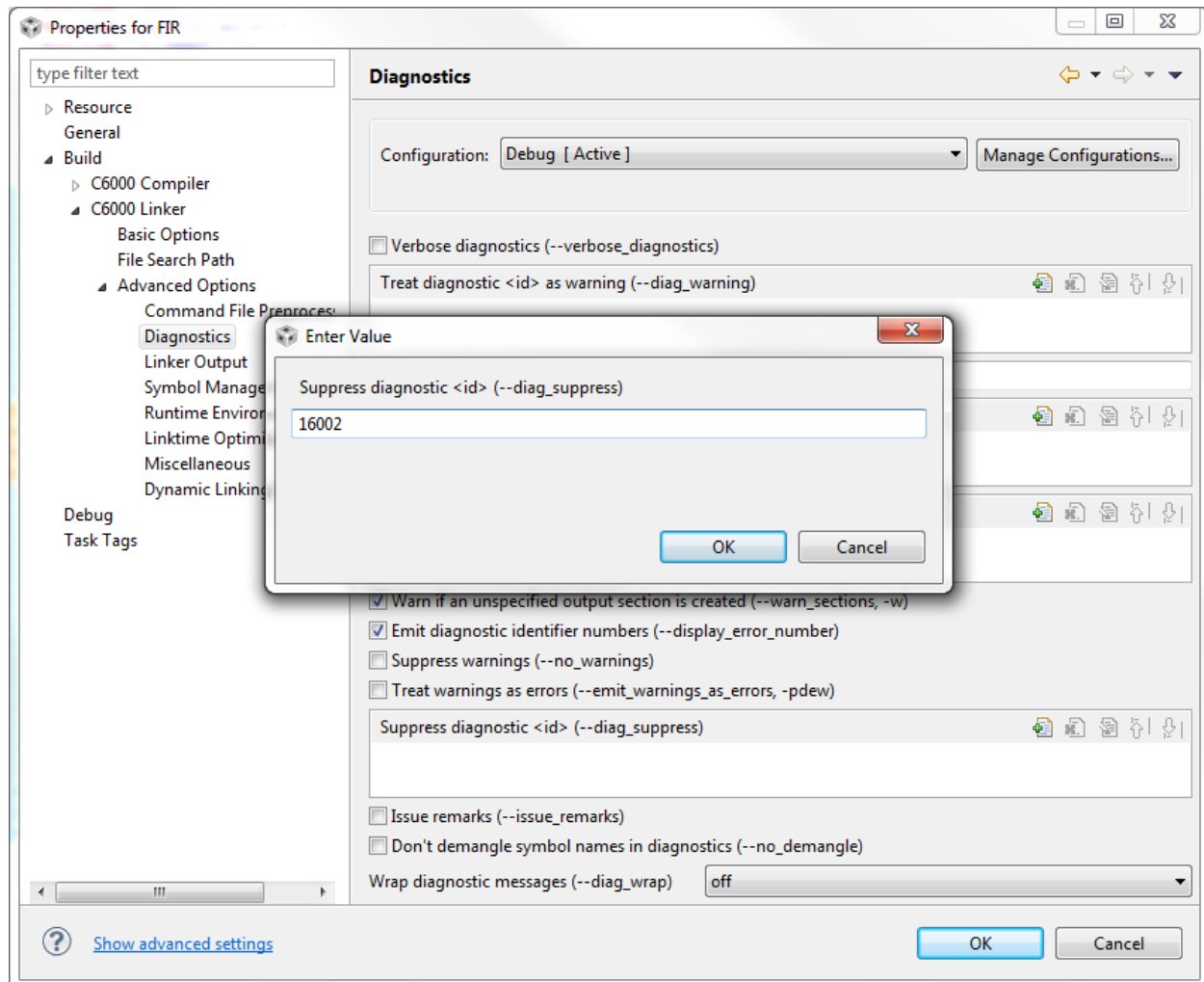
Part 7: Define stack and heap size

- This step is not necessary, but will eliminate warning messages
- Expand the C6000 Linker menu and select “Basic Options”
- Set stack and heap sizes both to 0x400



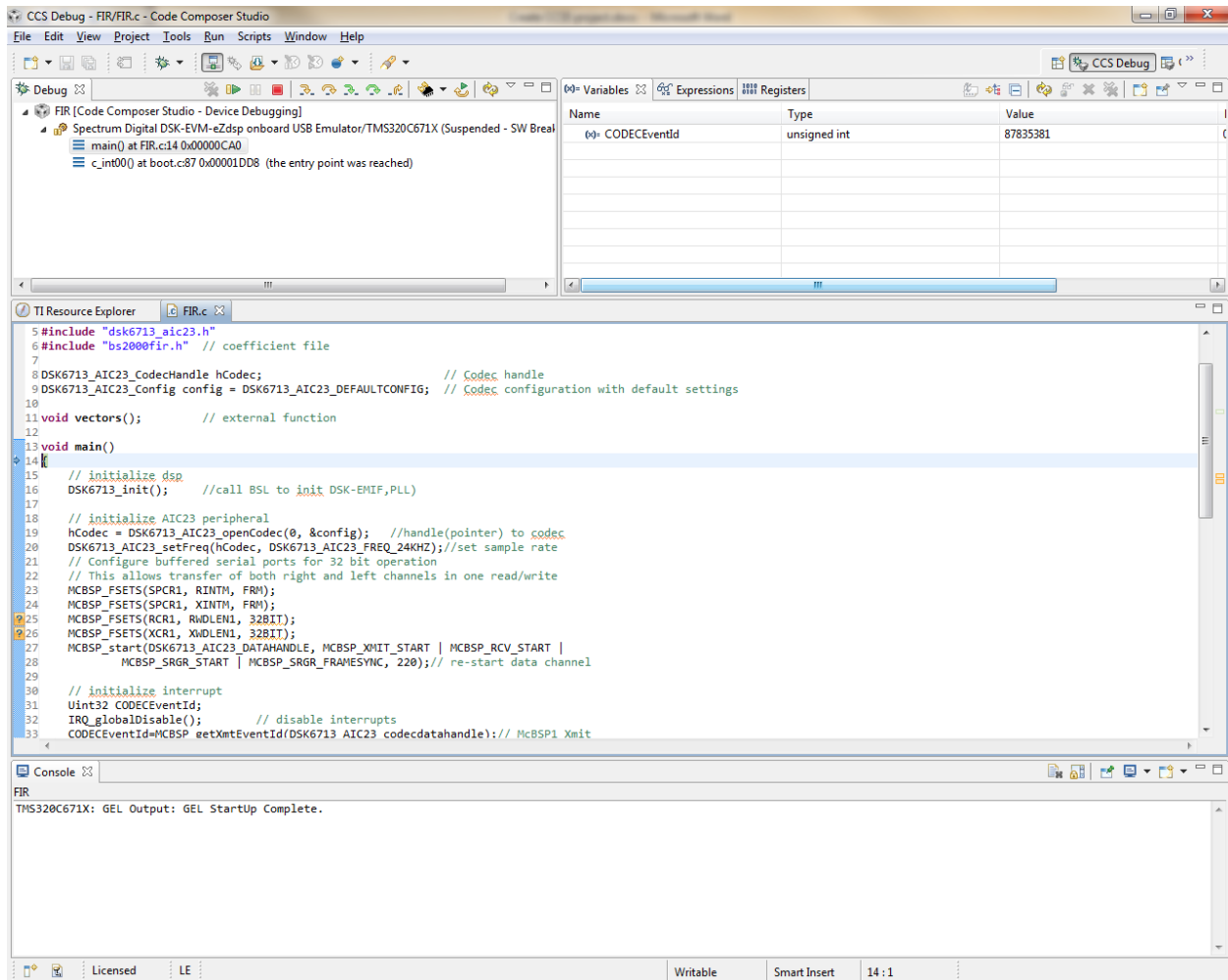
Part 8: Suppress “vendor section” warning

- Old library versions do not have the vendor section build attribute. This is used for verifying build compatibility, but is not required.
- Expand the C6000 Linker → Advanced Options menu. Select “Diagnostics”
- Set Suppress Diagnostic <id> to “16002” to ignore this warning



Part 9: Build and run the project

- Pull down the Project menu and select “Build All”
- If the project builds without errors, you can run the code on the DSP by pulling down the Run menu and selecting “Debug”



- Note: Selecting Debug will also build the project before running it
- You may use this project as a template for new DSK6713 projects. Simply copy and paste the project within the Project Explorer window. You will be prompted for a new project name.